Exam Solution for TDT4186 Resit

Spring 2025

1 Multi-Choice Questions

Automatic grading. Attention: Each question is worth 1.5 points.

1. Monolithic Kernel
2. User mode and kernel mode
3. A program in execution
4. Running to Blocked
5. createprocess()
6. To pause the execution of the parent process until a child process finishes
7. A process is a program in execution, while a thread is a subset of a process.
8. Stack
9. SJF
10. RR
11. Priority scheduling and RR
12. When the process exceeds its time quantum in the current queue
13. An address generated by the CPU that is mapped to a physical address
14. It allocates fixed memory blocks to processes.
15. Segmentation eliminates all external fragmentation.
16. Worst-Fit
17. 1,4
18. 1,2
19. The page size can be different for each process.

20. 15 bits
21. 36
22. The TLB stores copies of frequently used page table entries.
23. Least-recently used
24. 512, 9 bits
25. Preemption
26. Condition variables store shared data between threads.
27. 1
28. Reduces CPU overhead by allowing devices to transfer data directly to memory
29. The I/O device signals the CPU when it has completed its task, suspending the current process if needed.
30. SSTF (Shortest Seek Time First)
31. Higher read/write speed
32. 11
33. Superblock
34. Filename
35. A small integer used by a process to access an open file or I/O resource
36. 0
37. It keeps track of free and used disk blocks.
38. $1 \rightarrow 3 \rightarrow 2 \rightarrow 4$

2 Multi-part questions

For multi-part questions, the automatic grading may fail to award some answers which are slightly different from the provided solutions or written in different phrases. Thus, it is worth noting that all wrong blanks should go through a scrutiny check. Answers that provide a close meaning can be awarded the point without deduction.

Process

The increment of the counter variable should be in the parent section and executed before calling exec() in the child.

- fork()
- exec()
- wait()
- A

Scheduling

- priority or weight
- O(1)
- P5
- larger
- 12

Paging

- 10A
- OEA
- 184
- E12
- 5ABCE12 (the final level page directory stores the frame number)

Concurrency

Explanation: The ticket dispenser is a variation of the producer-consumer problem. The customer thread is the producer, while the bank_staff thread is the consumer. Thus, the initializations of the three semaphores are the same as in the producer-consumer problem. Both the bank_staff thread and the consumer threads access the waiting queue, while the bank_staff also has its own small queue — the service desk — which can be considered as a queue with a single empty slot.

1. A customer first needs to check whether an empty seat is available, so it waits on the empty semaphore.
2. The customer grabs the lock to request a ticket. After the ticket is successfully grabbed, the customer releases the lock.

3. Then, it updates the full semaphore.
4. In the end, the customer waits on the bank staff's service.

Answers:

- semaphore empty = 5;
- semaphore full =0;
- sem_wait(empty)
- sem_post(mutex)
- sem_post(full)
- sem_wait(service)

File systems

As long as your answers convey the same meaning, you will receive the points.

- To delete a file, the OS performs an **unlink** by removing the file's name and decreases **the reference count** (link count) of the inode. If the reference count reaches zero, the OS performs garbage collection to automatically reclaim the inode.